

ITA06 - Machine Learning

Exp 2: For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

Aim

To implement and demonstrate the **Candidate-Elimination Algorithm** for finding the version space (all hypotheses consistent with the training examples) from a given CSV file.

Algorithm Steps

1. Read training data from a CSV file.
2. Initialize:
 - Specific hypothesis (S) with the first positive example.
 - General hypothesis (G) with the most general hypotheses.
3. For each training example:
 - If positive:
 - Generalize S minimally.
 - Remove inconsistent hypotheses from G.
 - If negative:
 - Specialize G minimally.
 - Remove inconsistent hypotheses.
4. Display the final Specific Boundary (S) and General Boundary (G).

PROGRam:

```
import pandas as pd
```

```
import numpy as np
```

```
# Load CSV file
```

```
data = pd.read_csv('trainingdata.csv')
```

```
print("Training Data:\n")
```

```
print(data)
```

```

# Extract attributes and target
concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])

# Initialize Specific Hypothesis
S = concepts[0].copy()

# Initialize General Hypothesis
G = [["?" for i in range(len(S))] for j in range(len(S))]

print("\nInitial Specific Hypothesis:")
print(S)

print("\nInitial General Hypothesis:")
print(G)

for i, h in enumerate(concepts):

    if target[i] == "Yes":

        for x in range(len(S)):
            if h[x] != S[x]:
                S[x] = '?'
                G[x][x] = '?'

    if target[i] == "No":

```

